

fixed so as to be close to, but not necessarily touching the inner skin. Heat from the cables is assumed to escape through the inner skin only. The conduit can be metal or plastic.

b) Reference methods B1, item 4 of Table A.52.3 (insulated conductors in conduit on a wooden wall) and **B2**, item 5 of Table A.52.3, (multi-core cable in conduit on a wooden wall):

Conduit mounted on a wooden wall so that the gap between the conduit and the surface is less than 0,3 times the conduit diameter. The conduit can be metal or plastic. Where the conduit is fixed to a masonry wall the current-carrying capacity of the cable or insulated conductors may be higher. This subject is under consideration.

c) Reference method C, item 20 of Table A.52.3 (single-core or multi-core cable on a wooden wall):

Cable mounted on a wooden wall so that the gap between the cable and the surface is less than 0,3 times the cable diameter. Where the cable is fixed to or embedded in a masonry wall the current-carrying capacity may be higher. This subject is under consideration.

NOTE 1 The term "masonry" is taken to include brickwork, concrete, plaster and the like (other than thermally insulating materials).

d) Reference method D1, item 70 of Table A.52.3, (multi-core cable in ducts in the ground) and **D2** (multi-core cables designed to be buried directly in the ground – refer to manufacturer's instructions):

Cables drawn into 100 mm diameter plastic, earthenware or metallic ducts laid in direct contact with soil having a thermal resistivity of 2,5 K·m/W and a depth of 0,7 m (see also B.52.3).

Cables laid in direct contact with soil having thermal resistivity of 2,5 K·m/W and a depth of 0,7 m (see also B.52.3).

NOTE 2 With cables laid in the ground it is important to limit the temperature of the sheath. If the heat of the sheath dries out the soil, thermal resistivity may increase and the cable becomes overloaded. One way of avoiding this heating is to use the tables for 70 °C conductor temperature even for cables designed for 90 °C.

e) Reference methods E, F and G, items 32 and 33 of Table A.52.3 (single-core or multi-core cable in free air):

A cable so supported that the total heat dissipation is not impeded. Heating due to solar radiation and other sources shall be taken into account. Care shall be taken that natural air convection is not impeded. In practice, a clearance between a cable and any adjacent surface of at least 0,3 times the cable external diameter for multi-core cables or 1 time the cable diameter for single-core cables is sufficient to permit the use of current-carrying capacities appropriate to free air conditions.

B.52.6.2 Other methods

a) Cable on a floor or under a ceiling: this is similar to reference method C except that the current-carrying capacity for a cable on a ceiling is slightly reduced (see Table B.52.17) from the value for a wall or a floor because of the reduction in natural convection.

b) Cable tray system: a perforated cable tray has a regular pattern of holes so as to facilitate the use of cable fixings. The current-carrying capacity for cables on perforated cable trays have been derived from test work utilizing trays where the holes occupied 30 % of the area of the base. If the holes occupy less than 30 % of the area of the base, the cable tray is regarded as unperforated. This is similar to reference method C.

c) Cable ladder system: this construction offers a minimum of impedance to the air flow around the cables, i.e. supporting metal work under the cables occupies less than 10 % of the plan area.

d) Cable cleats, cable ties: devices for fixing cables to cable tray or bundling cables together

e) Cable hangers: cable supports which hold the cable at intervals along its length and permit substantially complete free air flow around the cable.

General notes to Tables B.52.1 to B.52.21.

NOTE 3 Current-carrying capacities are tabulated for those types of insulated conductor and cable and methods of installation which are commonly used for fixed electrical installations. The tabulated capacities relate to continuous steady-state operation (100 % load factor) for d.c. or a.c. of nominal frequency 50 Hz or 60 Hz.

NOTE 4 Table B.52.1 itemizes the reference methods of installation to which the tabulated current-carrying capacities refer. It is not implied that all these items are necessarily recognized in national rules of all countries.

NOTE 5 For convenience where computer-aided installation design methods are employed, the current-carrying capacities in Tables B.52.2 to B.52.13 can be related to conductor size by simple formulae. These formulae with appropriate coefficients are given in Annex D.

f) Cables in a ceiling: this is similar to reference method A. It may be necessary to apply the correction factors due to higher ambient temperatures that may arise in the junction boxes and similar mounted in the ceiling.

NOTE 6 Where a junction box in the ceiling is used for supply to a luminaire, the heat dissipation from the luminaire may provide higher ambient temperatures than prescribed in Tables B.52.2 to B.52.5, see also 522.2.1. The temperature may be between 40 °C and 50 °C, and a correction factor according to "Table B.52.14" has to be applied.